

VISHAL KHARBANDA

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SUMMARY

Backend Software Engineer with **3.5+ years** of experience building **Python-based microservices** and automation systems. Strong focus on REST APIs, CI/CD pipelines, Docker, and Linux-based deployments. Experienced in improving system performance and reliability in production environments, with working knowledge of Node.js and cloud platforms (AWS, Azure).

TECHNICAL SKILLS

Core Backend: Python, Node.js, Java, Microservices, REST APIs, Flask, FastAPI, Express, SQL, OOP, MVC, Unit Testing

DevOps & Cloud: GitHub Actions (CI/CD), Docker, NGINX, AWS, Azure, Zoho Server Management, PM2, Log Analysis

Databases & Data: MongoDB, RethinkDB, SQL Server, ArangoDB, PostgreSQL, ETL, Data Warehousing

Tools & Monitoring: Git, VS Code, Postman, SSH, Remote Monitoring, Structured Logging, Agile/Scrum

PROFESSIONAL EXPERIENCE

Tech Mahindra, Noida — Software Engineer

Dec 2023 – Present

- Designed and maintained scalable **Python-based microservices** for a mobile device testing platform, supporting distributed systems and automated interaction across Linux, Windows, and iOS environments.
- Built a specialized Python service that **reduced device battery consumption by 35%**, improving system performance, resource utilization, and remote device management reliability.
- Created and optimized **RESTful APIs** in Python (Flask/FastAPI) and Node.js (Express), enabling efficient data flow and real-time control of remote devices.
- Implemented **CI/CD pipelines** using GitHub Actions for automated builds, testing, and deployments; integrated code obfuscation workflows to ensure IP protection and secure releases across NGINX-based Zoho servers.
- Engineered secure encryption and obfuscation mechanisms across Python and Node.js services, achieving **100% compliance** with internal data protection and client security standards.
- Optimized data transformation and parsing pipelines using Python to process low-level device packet data from Qualcomm chipsets, **improving data pipeline efficiency by 40%**.
- Operated, monitored, and troubleshooted distributed backend services across **80+ production servers**, using PM2, SSH, logging, and monitoring tools to ensure high availability and system stability.

XenonStack, Mohali, India — Data Engineer

Aug 2022 – Sep 2023

- Designed and implemented **real-time ETL pipelines** using Debezium and Azure services, reducing data latency and improving system availability.
- Developed a **UPI fraud detection system** leveraging ArangoDB for secure and efficient pattern-based anomaly detection.
- Built and integrated Power BI dashboards with Azure Synapse and SQL Pool, **boosting analytics performance by 20%** and enabling faster, data-driven insights.
- Collaborated with cross-functional teams to streamline data ingestion and quality processes, **reducing integration time by 20%** and enhancing overall data reliability.

PROJECTS

TMDC (Tech Mahindra Device Cloud) — Tech Mahindra, Noida

Jan 2024 – Present

- Developed and enhanced a **mobile device cloud platform** enabling remote access and testing of real devices across Linux, Windows, and iOS environments.
- Built backend **Python microservices** for device orchestration, automation, and telemetry collection, following microservice architecture principles.

- Designed and integrated **RESTful APIs** to enable secure and low-latency communication between cloud services and physical devices.
- Improved platform reliability and observability by integrating structured logging, monitoring, and alerting across distributed services.

UPI Fraud Detection System — XenonStack, Mohali

Jan 2023 – Sep 2023

- Developed a **real-time fraud detection system** for UPI transactions using ArangoDB and pattern-based anomaly detection techniques.
 - Built **event-driven ETL pipelines** using Debezium and Azure services to enable near real-time transaction monitoring.
 - Architected optimized data models to identify suspicious behavioral patterns, improving fraud detection accuracy and response time.
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CERTIFICATIONS

- **Generative AI for Beginners** — Udemy (Aakriti E-Learning Academy), May 2026
Covered foundational concepts of Generative AI including prompt engineering, LLMs, and AI-driven automation.
 - **AI White Belt** — Tech Mahindra (SPOT Certificate)
Recognized for demonstrated AI technology proficiency and commitment to AI-first strategy; awarded for applied AI skills in enterprise workflows.
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EDUCATION

Bachelor of Technology (B.Tech.) — Computer Science Engineering

I.K. Gujral Punjab Technical University, Jalandhar | Chandigarh Group of Colleges, Landran

Graduated May 2022 | GPA: 8.2

Relevant Coursework: Software Designing and Programmable Logic, Advanced SQL